Energy Economics 150 (2025) 108839

Contents lists available at ScienceDirect

Energy Economics

journal homepage: www.elsevier.com/locate/eneeco

Dynamics of co-bubble networks across commodity futures prices and portfolio performance

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ARTICLE INFO

Keywords:
Co-bubble network
Commodity futures
Bubble transmission
COVID-19 outbreak and conflict between Russia and Ukraine
Portfolio implications

ABSTRACT

This paper introduces a price co-bubble network model then investigates the transmission of co-bubbles across 36 commodity futures prices, enabling a comprehensive understanding of the dynamic changes in co-bubble dependencies at both the overall and sector levels and during crisis periods. The analysis highlights several key results. Firstly, commodity futures within the Energy group demonstrate a higher propensity for experiencing co-bubbles compared to the Metals, Agricultural, and Grain and Cereal groups. Secondly, co-bubble occurrence is heterogeneous among pairs of commodity futures prices within different groups. Thirdly, the influence of co-bubble centrality is time-varying, reflecting its sensitivity to crisis periods. During the COVID-19 pandemic, Metals, particularly Gold and Palladium, exhibit a gradual improvement in their co-bubble network centrality ranking. During the Russo-Ukrainian war, Carbon Emissions maintain a dominant position in the centrality ranking. Fourthly, a portfolio analysis demonstrates that commodity portfolios constructed based on the results of the proposed co-bubble network outperform the baseline strategy, yielding higher Sharpe ratios and cumulative returns. These results extend our understanding on the dynamics of co-bubble dependencies in the commodity markets, and highlight the potential benefits of incorporating co-bubble dynamics into the portfolio construction.

1. Introduction

Commodities as an asset class have gathered widespread recognition among various economic actors over the past two decades (Christoffersen et al., 2019). Significant attention is given to the dynamic behaviour of commodity prices, particularly during crisis periods such as the 2008 global financial crisis (Zhang and Broadstock, 2020), the COVID-19 outbreak, and the Russian-Ukrainian war (Wang et al., 2022; Bouri et al., 2023). Importantly, the commodity market is heterogeneous, covering various commodity groups such as Energy, Metals, Agricultural, and Grains, which play important role in the global economy. Commodity price developments are crucial for economic activity and growth, inflation, poverty, food security, and climate change. Specifically, energy commodities such as crude oil and natural gas play a key role in the production process of modern societies and their price shocks often raise significant macroeconomic c, without ignoring the high carbon footprints of such fossil fuels. Metals are central to the global economy through their intermediate input role in industrial production and construction, which is especially the case of industrial metals. As for precious metals, they are mainly regarded as potential safe-haven assets during turmoil periods. Agricultural and grains commodities are detrimental for economic transformation, food security and nutrition. Commodity sectors are diverse, offering diversification and hedging opportunities (Rehman et al., 2019; Abid et al., 2020; Nekhili et al., 2021). However, the relationships between commodities tend to vary over time and intensify during crisis periods (Zhang and Broadstock, 2020; Bouri et al., 2021; Farid et al., 2022), challenging diversification potentials and hedging strategies. The prevailing literature in mainstream economics and finance of commodities also considers price characteristics (Pindyck, 2001) return interdependence (Kang et al., 2017; Ji et al., 2018; Chen et al., 2021; Bouri et al., 2023), volatility interdependence (Nazlioglu et al., 2013; Barbaglia et al., 2020; Bouri

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https://doi.org/10.1016/j.eneco.2025.108839
Received 21 August 2023; Received in revised form 28 June 2025; Accepted 12 August 2025
Available online 18 August 2025

et al., 2021, 2023), jumps (Diewald et al., 2015; Sévi, 2015; Nguyen and Prokopczuk, 2019) and co-jumps (Maslyuk-Escobedo et al., 2017; Semeyutin et al., 2021; Semeyutin and Downing, 2022; Zhang et al., 2024). The long-term cyclical pattern of commodity prices is well-known (Erten and Ocampo, 2013; Rossen, 2015), and recently Bouri et al. (2023) support the strong positive correlations among various commodities in the long-run, while highlighting that commodity market integration evolves over time and is influenced by crises.

The prices of many commodities are affected not only by demand/supply fundamentals,[1] but also by non-macroeconomic fundamentals such as weather conditions (D'Agostino and Schlenker, 2016; Zhang et al., 2023), government policies, geopolitical risk (Tiwari et al., 2021), and hedging and speculation (Ji et al., 2020). Accordingly, they can be subject to extreme price fluctuations, particularly under extreme events, potentially resulting in price bubbles. The existing literature examines the formation of bubbles in commodities belonging to various groups, such as Metals (Maghyereh and Abdoh, 2022; Wang et al., 2023), Agricultural (Gutierrez, 2013; Etienne et al., 2014; Chen et al., 2023), and Energy (Bohl et al., 2013; Gronwald, 2016; Caspi et al., 2018; Su et al., 2017; Liu and Lee, 2018; Sharma and Escobari, 2018; Akcora and Kocaaslan, 2023). Price bubbles in some commodity markets are driven by news-based economic sentiment (Maghyereh and Abdoh, 2022) and geopolitical risk and speculation (Wang et al., 2023).

Previous studies also focus on the interdependence and co-movement between commodity prices (Zhang and Broadstock, 2020; Bouri et al., 2023; Wang et al., 2022), highlighting the importance of financialization (Tang and Xiong, 2012), sentiment (Maslyuk-Escobedo et al., 2017; Ji et al., 2020), and extreme events and crisis periods (Bouri et al., 2023; Iqbal et al., 2023).[2] The commodity literature considers co-bubbles[3] in a limited way. El Montasser et al. (2023) examine co-bubbles between Brent oil prices and agricultural commodity prices based on a vector autoregressive model by Nielsen (2010), but do not pay attention to the transmission of co-bubbles, do not consider a network model capable of illustrating the co-bubble dependency and intensity, and also overlook the portfolio implications for the entire universe of commodity futures. Furthermore, the methods applied by El Montasser et al. (2023) only consider the co-bubbling between two commodities, lacking a comprehensive analysis of the transmission of co-bubbles among multiple pairs of commodities. Taken together, the connections between co-bubbles in commodity prices have received limited attention in the commodity literature, both at the aggregate level and within each commodity group, and the potential portfolio implications are overlooked.

In light of the above-mentioned shortcoming and gaps in the related literature, it is important to offer investors, portfolio managers, and scholars not only a comprehensive analysis of the co-bubble transmission across a wide range of commodities belonging to various commodity sectors (Energy, Metals, Agricultural, and Grain and Cereal) within a co-bubble network model, but also the portfolio implications of the significant transmission of co-bubbles. In this regard, fully grasping this intricate transmission of co-bubbles is a challenging endeavour due to the complex and diverse interactions that govern it. The occurrence of extreme events in each respective commodity market often leads to significant and unforeseen fluctuations in the prices of commodities and possibly the formation of bubbles or co-bubbles. Interestingly, a co-bubble network analysis can capture and reveal how commodity futures prices co-bubble and how the dynamic nature of their co-bubble behaviour evolves over time and during crisis periods. As for the portfolio implications arising from the presence of co-bubbles in commodity markets, they can be assessed in light of the results of proposed co-bubble network model. Notably, they should be appreciated by investors and portfolio managers, especially in the presence of extreme events and crisis periods. Furthermore, they are potentially crucial for the design of refined investment strategies. For policymakers, co-bubbles and the transmission of co-bubbles can have an impact on market stability and might concern commodity importers, which could further accentuate the importance of our current analysis.

Accordingly, our current study address several questions related to identifying which commodity futures serve as bridges for cross-industry group bubbles, determining the highest-ranking commodity futures in the co-bubble network during normal and crisis periods, understanding how the dynamics of interrelated bubbles in commodity market prices change during crisis events, and uncovering the portfolio implications of significant co-bubbles in commodity markets. Specifically, our study identifies the co-bubble transfer mechanism in 36 commodity futures prices through the implementation of co-bubble network models and examines their portfolio implications. Methodologically, we follow a three-step approach. Firstly, we date-stamp price bubbles in various commodities by employing the backward supremum augmented Dickey-Fuller (BSADF) tests proposed by Phillips et al. (2015a, 2015b). Secondly, we detect bubbles and co-bubbles using the approach introduced by Bouri et al. (2019). Finally, we construct the co-bubble network, inspired by the work of Ding et al. (2024), to investigate co-bubble dependencies in commodity futures. We do this for the overall commodity market and for each commodity group using a network analysis, while capturing the dynamic changes in co-bubble dependencies over time and across crisis periods.

Unlike traditional co-bubble methods, such as those employed by Bouri et al. (2019) in cryptocurrencies and El Montasser et al. (2023) in energy and agricultural commodities, the co-bubble network model we propose in this paper can simultaneously consider the bubble dependency among multiple pairs of commodity futures. Interestingly, this comprehensive model can reveal the co-bubble relationships between commodity futures prices and provide a holistic understanding of commodity market dynamics over time. Furthermore, the co-bubble network model allows us to identify the transmission paths of co-bubbles across all commodities and within commodity groups, providing a more refined understanding of price co-bubble transmission and practical insights into the portfolio implications of co-bubbles in various commodity groups.

The main contributions of this study can be summarized as follows: Firstly, unlike previous studies considering the formation of commodity bubbles in the energy sector such as crude oil (Caspi et al., 2018; Su et al., 2017) and natural gas (Akcora and Kocaaslan, 2023), the metal sector such as precious metals (Maghyereh and Abdoh, 2022) and nickel (Wang et al., 2023), and the agricultural and grain sectors (Etienne et al., 2014; Chen et al., 2023), we propose a co-bubble network model to investigate the co-bubble dependency among 36 commodity futures prices, enabling a comprehensive understanding of how co-bubbles are transmitted across commodities and how they evolve over time. Such a model allows us to identify key nodes in commodity futures, shedding light on the commodity markets with the most significant impact on commodity bubbles. In contrast to the approach taken by El Montasser et al. (2023) to study the co-bubbling effect between energy and agricultural commodities, we assess the intensity of co-bubbles between a large set of commodity futures using a co-bubble network, which provides insights into which commodity futures exert a stronger influence through co-bubbles and which commodities are more susceptible to the

[1] Chan et al. (2024) consider monetary policy responses to the formation of bubbles in clean and brown sectors, highlighting the role of supply-side and demand-side shocks for the formation of bubbles.

[2] On a related front, previous studies consider the crude oil-bank stock nexus (Zhang et al., 2024a), crude oil-energy stock nexus and the role of uncertainty (Zhang et al., 2024b), and the effect of climate risk on market integration of Chinese energy firms (Zhang et al., 2024c).

[3] Some studies present compelling evidence supporting the existence of co-bubbles in various asset prices. Bouri et al. (2019) examines bubbles and co-bubbles in seven cryptocurrencies (Bitcoin, Ripple, Ethereum, Litecoin, Nem, Dash, and Stellar) and find that price bubbles in one cryptocurrency can trigger price bubbles in other cryptocurrencies. Similar results are observed by Haykir and Yagli (2022). Furthermore, Aloosh et al. (2022) explore co-bubbles between stocks restricted by Robinhood and cryptocurrencies.

prices co-bubbles of other commodities. Secondly, we compare and analyse the similarities and differences between the co-bubble networks of the entire commodity markets and each commodity group. This allows us to gain insights into how co-bubble relationships differ within various commodity groups. Thirdly, we study the time-varying nature of co-bubbles, which is useful to capture how co-bubble relationships change over time and under extreme events. Specifically, we compare the structure of the co-bubble networks during the COVID-19 crisis and the Russo-Ukrainian conflict, providing an assessment of the diverse co-bubble impacts of these crises on commodity futures. In fact, such a dynamic analysis helps us understand how global health and geopolitical events can influence the time-evolution of co-bubble transmission in commodity markets. Finally, we consider the portfolio implications of co-bubbles by constructing enhanced commodity portfolios, which extends the findings of previous studies such as El Montasser et al. (2023) by revealing how the co-bubble network can be exploited in portfolio management. By considering the presence of co-bubbles in the construction of commodity portfolios, we provide insights into how portfolio managers can better manage the risks associated with co-bubbles and enhance the performance of their commodity portfolios.

Our empirical analysis reveals the presence of co-bubbles in commodity futures prices, with some evidence of heterogeneity across various commodities. Co-bubbles are generally more prominent within each commodity group, with the Energy group showing a higher likelihood of experiencing co-bubbles. The impact of co-bubble centrality is time-varying, showing its sensitivity to crisis periods. Notably, the dynamics of co-bubble influence rankings experience significant changes during the COVID-19 pandemic and Russo-Ukrainian war. During the pandemic, there is a gradual improvement in the ranking of the Metals group, in particular, for Gold and Palladium, in terms of co-bubble network centrality. During the Russo-Ukrainian war, Carbon Emissions maintain its dominant position in the centrality ranking. Additional analysis of the portfolio implications of the main findings reveals that employing mean-variance portfolio strategies based on the dynamic ranking of co-bubble influence across commodities can result in higher Sharpe ratios compared to the baseline strategy.

Moving forward, Section 2 describes the bubble and co-bubble tests, as well as the co-bubble network model. Section 3 provides the empirical results and their discussion. Section 4 presents a portfolio analysis, investigating the implications of co-bubble dynamics for portfolio managers. Finally, Section 5 concludes the paper.

2. Methods

In this section, we examine the dynamics of bubbles and co-bubbles of 36 commodity futures over the sample period. After detecting bubbles in every commodity under study, we propose pairwise co-bubble scores to measure the bubble similarity between two commodities. Then, we leverage these co-bubble scores to build matrices and accordingly co-bubble networks. The analysis is conducted in static and time-varying settings.

We start by introducing an asset pricing model with a rational bubble. Following Sharma and Escobari (2018), we define a conceptual framework for commodity prices bubbles:

$$P_{t+1} \equiv \frac{E_t(P_{t+1} + C_{t+1})}{1+R}, \tag{1}$$

where P_{t+1} and C_{t+1} are the commodity price and the convenience yield from period t to $t+1$, respectively; R is the constant discount rate.

Following Campbell and Shiller (1988) and Sharma and Escobari (2018), Eq. 1 can be written with a log-linear approximation:

$$p_t = f_t + b_t \tag{2}$$

where, the logarithm of the commodity price, $p_t = \log(P_t)$, can be explained by two components; f_t denotes the fundamental component, and b_t denotes the rational bubble component. Campbell and Shiller (1988) and Sharma and Escobari (2018) derive two components as follows:

$$f_t = \frac{\kappa - \gamma}{1-\rho} + (1+\rho)\sum_{i=0}^{\infty} \rho^i E_t(c_{t+1+i}) \tag{3}$$

$$b_t = \lim_{i\to\infty} \rho^i E_t(p_{t+i}) \tag{4}$$

where,$c_t \equiv \log(C_t)$,$\gamma \equiv \log(1+\mathrm{R})$, $\rho \equiv 1/\left[1+e^{(\overline{c-p})}\right]$, $\overline{c-p}$ is the average convenience yield–price ratio, and k is given by:

$$\kappa = -\log(\rho) - (1-\rho)\log\left(\frac{1}{\rho} - 1\right). \tag{5}$$

2.1. Detecting bubbles and pairwise co-bubble scores

In the previous subsection, we briefly describe how to isolate the bubble component from the asset pricing model. In this subsection, we mainly focus on the bubble tests. Phillips et al. (2015a) propose the generalized supremum augmented Dickey–Fuller (GSADF) test to identify multiple explosiveness periods (Long et al., 2016; Corbet et al., 2018), which is a robust technique and extends the supremum augmented Dickey–Fuller (SADF) test of Phillips et al. (2011). The SADF test has the following recursive regression:

$$y_t = \mu + \beta y_{t-1} + \sum_{i=1}^{p} \delta_{r_\omega} \Delta y_{t-i} + \varepsilon_t, \tag{6}$$

where y_t is the commodity future price; μ and p are the intercept and the maximum number of lags, respectively; δ denotes the differenced lags coefficients; r_ω denotes the rolling interval window size of the regression, defined by $r_\omega = r_2 - r_1$. Testing for a bubble in commodity future price is based on a right-tail variation of the standard ADF unit root test, and the null hypothesis is $H_0 : \beta = 1$ against explosive bubbles, $H_1 : \beta > 1$. The SADF statistic is defined as:

$$SADF(r_0) = Sup_{r_2\in(r_0,1)} ADF_{r_2} \tag{7}$$

To allow more flexible estimation windows, Phillips et al. (2015a) propose the generalized GSADF test, which can address the potential sensitivity of the SADF test. The GSADF statistic is defined as:

$$GSADF(r_0) = Sup_{r_2\in(r_0,1)} SADF_{r_2}(r_0) \tag{8}$$

As a date-stamping strategy, the GSADF can consistently estimate the start and end of bubbles for commodity futures, and Phillips et al. (2015b) propose the backward SADF (BSADF) as follows:

$$BSADF(r_0) = Sup_{r_1\in(0,r_2-r_0)} ADF_{r_1}^{r_2} \tag{9}$$

where, r_2 denotes the end of the rolling interval window, and $r_2 - r_0$ denotes the window size. The estimates of bubble periods are defined based on the GSADF test by:

$$\widehat{r_e} = inf_{r_2\in(r_0,1)}\left\{r_2 : BSADF_{r_2} > cv_{r_2}^{\alpha_T}\right\} \tag{10}$$

$$\widehat{r_f} = inf_{r_2\in(\hat{r}_e,1)}\left\{r_2 : BSADF_{r_2} > cv_{r_2}^{\alpha_T}\right\} \tag{11}$$

where, $cv_{r_2}^{\alpha_T}$ denotes the 100 (1–α_T)% critical value of the Sup ADF statistics based on r_2 observations, and α_T denotes a constant value of 5 %. The initial sample size $r_0 = 0.01 + \frac{1.8}{\sqrt{T}}$, and all simulations are constructed based on 1999 replications.

Based on the approach introduced by Bouri et al. (2019), we denote the cumulative number of bubbles detected for commodity future i from day 1 to day t as:

$$B_{i,t} = \sum_{1 \le d \le t} 1_{\left\{BSADF_{r_2,t} > cv_{r_2,t}^{\alpha_T}\right\}} \tag{12}$$

where {1} is the indicator function. We define co-bubbles between any pair of commodity futures prices as the simultaneous occurrence of bubbles in the pair of commodity futures prices, i.e. the intersection of their individual bubbles. We denote the cumulative number of co-bubbles between commodity futures i and commodity futures j from day 1 to day t as:

$$CB_{ij,t} = \sum_{1 \le d \le t} 1_{\left\{BSADF_{i,r_2,t} > cv_{i,r_2,t}^{\alpha_T}, BSADF_{j,r_2,t} > cv_{j,r_2,t}^{\alpha_T}\right\}} \tag{13}$$

The co-bubble indices between pairs of commodity futures have three useful characteristics. Firstly, they are non-negative, where higher values indicate a stronger co-bubble relationship. Secondly, the co-bubble indices are scaled, allowing for comparison between co-bubble relationships among pairs of commodity futures. Thirdly, the co-bubble indices are symmetric, reflecting the symmetry of the co-bubble relationship between two commodity futures.

2.2. Co-bubble networks

Traditional co-bubble methods only consider co-bubbles between two financial assets, thus they cannot simultaneously consider the bubble dependencies among multiple commodity futures. Motivated by the intuition that commodity futures are more inter-dependent if they co-bubble together more often, we propose pairwise co-bubble scores to measure the bubble similarity between commodity futures.

Using the pairwise co-bubble measures, we construct the corresponding $N \times N$ co-bubble matrix C(T) for commodity futures as follows:

$$(C(T))_{ii} = \frac{B_i(T)}{T} \text{ for } 1 \le i \le N, \text{ and } (A(T))_{ij} = \frac{CB_{ij}(T)}{T} \text{ for } 1 \le i \ne j \le N. \tag{14}$$

There are several aspects of the co-bubble network which can extend our understanding of co-bubbles in commodity futures markets. Firstly, the co-bubble network helps us identify which commodity futures tend to experience more co-bubbling behaviour. This allows us to understand the bubble interrelationships and interdependencies across commodity futures. Secondly, the co-bubble network analysis can shed light on the efficiency of the commodity future markets. This is based on the rationale that if co-bubbles occur frequently and persistently, they may indicate inefficiencies or trading opportunities which can be exploited by market participants. Finally, the co-bubble network can be used to construct optimal commodity futures portfolios. For example, by considering the centrality ranking of co-bubble influence between commodity futures, investors and portfolio managers can construct portfolios that are better diversified and resilient to co-bubble events.

The co-bubble network represents complex structures in commodity futures and has a symmetrical matrix. We consider dynamic co-bubble networks of commodity futures, $G_t = (V_t, E_t), t \in T$, where each vertex $N_{i,t} \in V$ represents a certain commodity future at time t, and a weighted edge $e_{i,j}(t) \in E$ denotes a type of co-bubble dependency between two commodity futures at time t; the weight being the corresponding co-bubble intensity.

3. Empirical results

3.1. Data

We collect daily closing prices against the USD for the 36 most common commodity futures from the Wind database over the period July 5, 2016, to July 1, 2023. Our sample period, although being dictated by the availability of price data on these numerous commodity futures, is comprehensive. It covers key events and crisis periods, notably the COVID-19 outbreak and the Russo-Ukrainian conflict, which have affected the supply and demand fundamentals and accordingly the market dynamics of several commodities. The resulting dataset contains a total of 1707 daily observations for each commodity. Table 1 presents descriptive statistics for the prices of all commodity futures. As our aim is to compute co-bubble scores between pairs of commodity futures, we date-match all data to ensure that there are identical trade dates. The 36 commodities are categorized into four sectors: Energy (7 commodities), Metals (10 commodities), Agricultural (11 commodities), and Grain and Cereal (8 commodities). Notably, we include carbon emission allowances as part of the Energy commodity sector due to its significance in decarbonisation efforts. Consistent with the approach of Bouri et al. (2023), we focus on futures prices instead of spot prices as the former play a more crucial and leading role in the price formation of commodities (evidence for energy can be found in Shrestha (2014), and for metals in Figuerola-Ferretti and Gonzalo (2010)). Regarding the gasoline commodity, we consider two futures prices: RBOB and London. This distinction arises from the fact that RBOB gasoline futures are primarily influenced by factors specific to the United States, while London gasoline futures are influenced by factors specific to the European market (as observed in Mirantes et al. (2012)). It is worth noting that there exist commodity futures of the same variety, and this is evident in the disparity of reported bubble results between RBOB and London for gasoline. For wheat, we consider two major trading hubs that provide valuable information on global wheat market trends and price discovery. Interestingly, we observe different bubble behaviours between the two types. Lastly, for coffee, we include both London and US coffee commodities as US coffee futures are based on Arabica coffee, while London coffee futures are based on Robusta coffee (as referenced in Thompson (1986)).

3.2. Results for commodity future bubbles

Table 2 presents the statistics for bubbles in commodity futures over the full sample period (July 5, 2016, to July 1, 2023). Lead, US Wheat, Platinum, Silver, and London Cocoa exhibit the lowest bubble activity, with 40, 43, 60, 85, and 122 bubbles recorded, respectively, representing 2.34 %, 2.52 %, 3.51 %, 4.98 %, and 7.15 % of observed days. Conversely, Carbon Emissions, Tin, Soybean Oil, London Wheat, and Heating Oil account for the highest number of bubbles, with 800, 583, 474, 440, and 437 bubbles, respectively, representing 46.89 %, 34.17 %, 27.78 %, 25.79 %, and 25.61 % of observed days. Appendix Fig. A1 plots explosive behaviour periods in the Metals group based on the GSADF test, while Appendix Figs. A2, A3, and A4 plot explosive behaviour periods in the Agricultural group, Energy group, and Grain and Cereal group, respectively. We note that these commodity futures exhibit significant variations in price and explosive behaviours, particularly among individual commodities and across groups of commodities. These findings demonstrate the recurring nature of bubbles in Energy commodities in particular, which mostly occur in the period after the COVID-19 outbreak and tend to exhibit some persistency.

3.3. Co-bubble network analysis I: all 36 commodity futures

Fig. 1 provides a comprehensive visualization of commodity futures co-bubble dependency by constructing a co-bubble network-based matrix, denoted by $B_{2016-2023}$, which includes all 36 commodities over the full sample period from July 5, 2016 to July 1, 2023. Each vertex in Fig. 1 represents an individual commodity. The colour distribution ranges from deep green to deep purple, reflecting the intensity of co-bubble dependencies between pairs of commodities.

Some interesting results emerge from Fig. 1. Firstly, commodities belonging to the Energy group show a higher probability of experiencing co-bubbles, which is reflected in the major green colours. Additionally, there exists a notable heterogeneity in the co-bubble dynamics among commodity futures, even within the same subgroup. For example, within the Metals group, Silver, Platinum and Lead have a deeper purple

Table 1
Summary statistics of commodity futures daily prices.

Commodity group	Individual commodity futures	Sample period	Mean	Max	Min	SD
Metals	**Silver**	July 5,2016-July 1,2023	19.59	29.418	11.77	3.96
Metals	**Palladium**	July 52,016-July 12,023	1587.64	3289	605.5	634.63
Metals	**Platinum**	July 52,016-July 12,023	951.64	1297.1	595.2	104.16
Metals	**Lead**	July 52,016-July 12,023	2116.14	2669	1585.5	214.97
Metals	**Copper**	July 52,016-July 12,023	3.2215	4.9	2.07	0.71
Metals	**Aluminium**	July 52,016-July 12,023	2114.49	3849	1462	412.81
Metals	**Zinc**	July 52,016-July 12,023	2796.91	4498.5	1815.5	467.91
Metals	**Tin**	July 52,016-July 12,023	22,840.88	48,650	13,250	6893.10
Metals	**Nickel**	July 52,016-July 12,023	16,111.05	81,051.5	8747.5	6007.18
Metals	**Gold**	July 52,016-July 12,023	1555.51	2069.4	1129.8	273.35
Agricultural	**London Cocoa**	July 52,016-July 12,023	1787.38	2722	1326	237.73
Agricultural	**London Coffee**	July 52,016-July 12,023	1770.44	2930	1121	366.89
Agricultural	**London Sugar**	July 52,016-July 12,023	439.26	720.1	294	98.96
Agricultural	**Lumber**	July 52,016-July 12,023	539.13	1686	259.8	254.94
Agricultural	**Orange Juice**	July 52,016-July 12,023	149.29	293.9	91.25	42.66
Agricultural	**US Sugar**	July 52,016-July 12,023	15.91	26.99	9.21	3.78
Agricultural	**US C Coffee**	July 52,016-July 12,023	144.46	258.45	88	41.09
Agricultural	**US Cotton**	July 52,016-July 12,023	80.94	158.02	48.41	18.58
Agricultural	**US Cocoa**	July 52,016-July 12,023	2423.91	3318	1780	280.81
Agricultural	**Live Cattle**	July 52,016-July 12,023	122.00	181.5	83.83	18.03
Agricultural	**Lean Hogs**	July 52,016-July 12,023	74.69	122.88	37.33	17.39
Energy	**RBOB Gasoline**	July 52,016-July 12,023	1.77	3.0618	0.83	0.46
Energy	**WTI Crude Oil**	July 52,016-July 12,023	61.90	123.7	−37.63	18.19
Energy	**Brent Crude Oil**	July 52,016-July 12,023	66.54	127.98	19.33	18.42
Energy	**London Gasoline**	July 52,016-July 12,023	614.52	1522.5	190.75	216.30
Energy	**Natural Gas**	July 52,016-July 12,023	3.36	9.647	1.48	1.53
Energy	**Heating Oil**	July 52,016-July 12,023	2.07	5.13	0.61	0.73
Energy	**Carbon Emissions**	July 52,016-July 12,023	36.16	98.01	3.98	28.85
Grain and Cereal	**London Wheat**	July 52,016-July 12,023	179.00	361	111.75	46.28
Grain and Cereal	**Oats**	July 52,016-July 12,023	338.46	807	158.5	134.25
Grain and Cereal	**Corn**	July 52,016-July 12,023	460.07	818.25	301.5	138.94
Grain and Cereal	**Rice**	July 52,016-July 12,023	12.97	22.06	9.12	2.49
Grain and Cereal	**Soybeans**	July 52,016-July 12,023	1120.06	1769	791	258.93
Grain and Cereal	**Soybean Oil**	July 52,016-July 12,023	41.80	81.96	25.52	15.07
Grain and Cereal	**Soybean Meal**	July 52,016-July 12,023	354.17	520.2	280	59.24
Grain and Cereal	**US Wheat**	July 52,016-July 12,023	586.34	1425.25	361	169.79

Notes: This table reports summary statistics of daily prices of the 36 most common commodity futures over the sample period July 5, 2016 to July 1, 2023. It includes, the mean of prices, standard deviation (SD), and largest and smallest price observations (denoted by max and min, respectively).

Table 2
Statistics of bubbles detected.

Industry group	Commodity future	Number of bubbles	% of days with bubbles	Industry group	Commodity future	Number of bubbles	% of days with bubbles
Metals	**Silver**	85	4.98 %	**Agricultural**	**US Cocoa**	133	7.79 %
Metals	**Palladium**	317	18.58 %	**Agricultural**	**Live Cattle**	261	15.29 %
Metals	**Platinum**	60	3.51 %	**Agricultural**	**Lean Hogs**	210	12.30 %
Metals	**Lead**	40	2.34 %	**Energy**	**RBOB Gasoline**	336	19.69 %
Metals	**Copper**	400	23.44 %	**Energy**	**WTI Crude Oil**	275	16.11 %
Metals	**Aluminium**	417	24.44 %	**Energy**	**Brent Crude Oil**	376	22.03 %
Metals	**Zinc**	316	18.52 %	**Energy**	**London Gasoline**	382	22.39 %
Metals	**Tin**	583	34.17 %	**Energy**	**Natural Gas**	150	8.79 %
Metals	**Nickel**	225	13.18 %	**Energy**	**Heating Oil**	437	25.61 %
Metals	**Gold**	387	22.68 %	**Energy**	**Carbon Emission**	800	46.89 %
Agricultural	**London Cocoa**	122	7.15 %	**Grain and Cereal**	**London Wheat**	440	25.79 %
Agricultural	**London Coffee**	294	17.23 %	**Grain and Cereal**	**Oats**	224	13.13 %
Agricultural	**London Sugar**	364	21.33 %	**Grain and Cereal**	**Corn**	301	17.64 %
Agricultural	**Lumber**	237	13.89 %	**Grain and Cereal**	**Rice**	166	9.73 %
Agricultural	**Orange Juice**	153	8.96 %	**Grain and Cereal**	**Soybeans**	383	22.45 %
Agricultural	**US Sugar**	141	8.26 %	**Grain and Cereal**	**Soybean Oil**	474	27.78 %
Agricultural	**US Coffee**	197	11.54 %	**Grain and Cereal**	**Soybean Meal**	219	12.83 %
Agricultural	**US Cotton**	316	18.52 %	**Grain and Cereal**	**US Wheat**	43	2.52 %

Note: This table presents the number of bubbles detected in the price series of the 36 commodity futures under study, and the percentage (%) of days with bubbles.

distribution, indicating their lower sensitivity to bubbles than the other commodity futures. However, Copper, Aluminium and Tin have a deeper green distribution, indicating a higher sensitivity to bubbles than the other commodity futures. Secondly, some under-the-radar commodity futures, e.g. Soybean Oil, Carbon Emissions, and Tin, have a higher co-bubble dependency (the main colour connecting these commodity futures is green, compared to other commodity futures connections).

We offer some potential explanations for the above results. Firstly, the bubble sensitivity of commodity futures prices may be related to their specific characteristics. For example, the Energy group is affected by a variety of factors besides supply and demand fundamentals, such as geopolitical risk (Wang et al., 2022) and macroeconomic factors (Schalck and Chenavaz, 2015), making them more susceptible to the

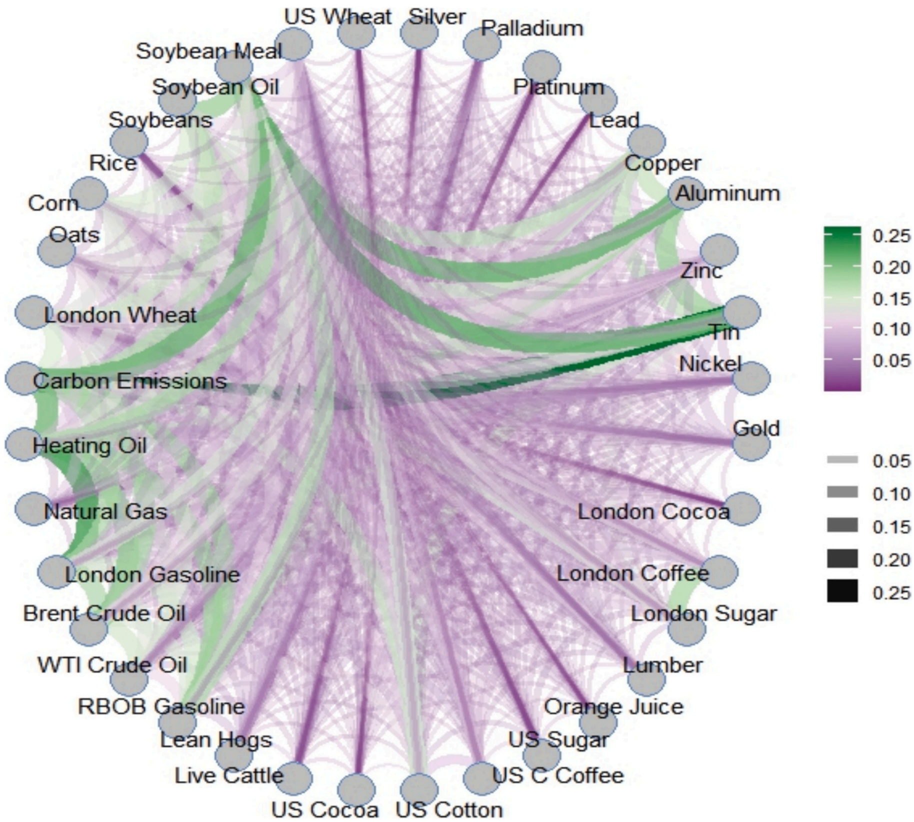

Fig. 1. Co-bubble network based on all 36 commodity futures. Notes: The edges connecting pairs of commodity futures represent the intensity of the co-bubble. In the colour bar (legend at bottom right), the colour distribution, ranging from deep green to deep purple, reflects the intensity of co-bubble relationships between pairs of commodity futures (deep green (deep purple) means higher (lower) intensity). In the colour bar (legend at top right), the colour distribution, ranging from light grey to black, reflects the width of the edge connecting the pairs of commodity futures, with grey (black) indicating a narrower (wider) width. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

emergence of co-bubbles. In addition, the more active trading activities in energy markets and the more rapid response of investors to price fluctuations may also increase the probability of co-bubbles. Secondly, there are information transmission and market linkages between commodity markets (Shahzad et al., 2021; Bouri et al., 2023; Iqbal et al., 2023), and co-bubbles between commodity futures may be affected by such linkage effects. For example, some commodity futures may be more susceptible to the transmission effect of price fluctuations in other commodity futures (for example, agricultural and energy commodities are interrelated through the biofuel expansion (see, Ji et al., 2018; Shahzad et al., 2021)), leading to a different probability and degree of co-bubble. Finally, differences in the availability and dissemination of information across commodity markets can also contribute to the patterns observed. Commodity futures that are more connected in terms of co-bubble dependency may have stronger information linkages, leading to synchronized price movements and higher susceptibility to bubbles.

We examine the bubble influence nodes based on all 36 commodity futures using the approach of Mantegna (1999a, 1999b) and filtering some trivial edges using a minimum spanning tree (MST), which helps maximize the sum of the co-bubble scores. Based on the work of Bonacich (1987), we employ eigenvector centrality to measure the influence of each commodity future on the co-bubble network. Fig. 2 shows the co-bubble network involving all 36 commodity futures based on MST. We note that all the commodity markets are polarized and have no distinctive features or patterns. However, some interesting results emerge from it. For example, Silver and US Cotton have the strongest price bubble influence, indicating that the bubble formation in some commodity futures is influenced by the bubble formation in Silver and US Cotton futures. In other words, once a bubble occurs in Silver and US Cotton futures, the probability of having a bubble in other commodity futures increases.

3.4. Co-bubble network analysis II: inter-industry groups

Here, we focus on the inter-commodity groups, by considering four co-bubble networks. This helps us capture co-bubble dependency within each commodity group over the full sample period. Fig. 3 displays the four co-bubble networks based on inter-commodity groups (Energy, Metals, Agricultural, and Grain and Cereal). Each vertex belonging to each commodity group represents an individual commodity future. The colour distribution in the subgraphs, ranging from deep green to deep purple, reflects the intensity of co-bubble relationships across commodities.

It is worth noting that investors generally perceive that commodity futures within the same commodity group are more susceptible to co-bubble behaviour, which is supported by previous studies (Maghyereh and Abdoh, 2022). This is primarily attributed to the higher similarity in the demand and supply dynamics of commodities belonging to the same group (Bouri et al., 2021), as well as the common factors influencing the price formation of these commodities (for example, extreme weather conditions for agricultural and grain commodities, see D'Agostino and Schlenker (2016)). The four co-bubble networks reveal some meaningful phenomena in various inter-commodity groups. In the Metals group, there is a strong co-bubble dependency between Tin, Aluminium, and Copper. Lead has the lowest bubble sensitivity to the bubbling behaviour of other Metals commodity futures. In the Agricultural group, London Sugar has the strongest bubble dependence on London Coffee. In the Energy group, Heating Oil is more susceptible to the bubble formation of other commodity futures, especially London Gasoline and Carbon Emissions, whereas Natural Gas has a relatively lower bubble

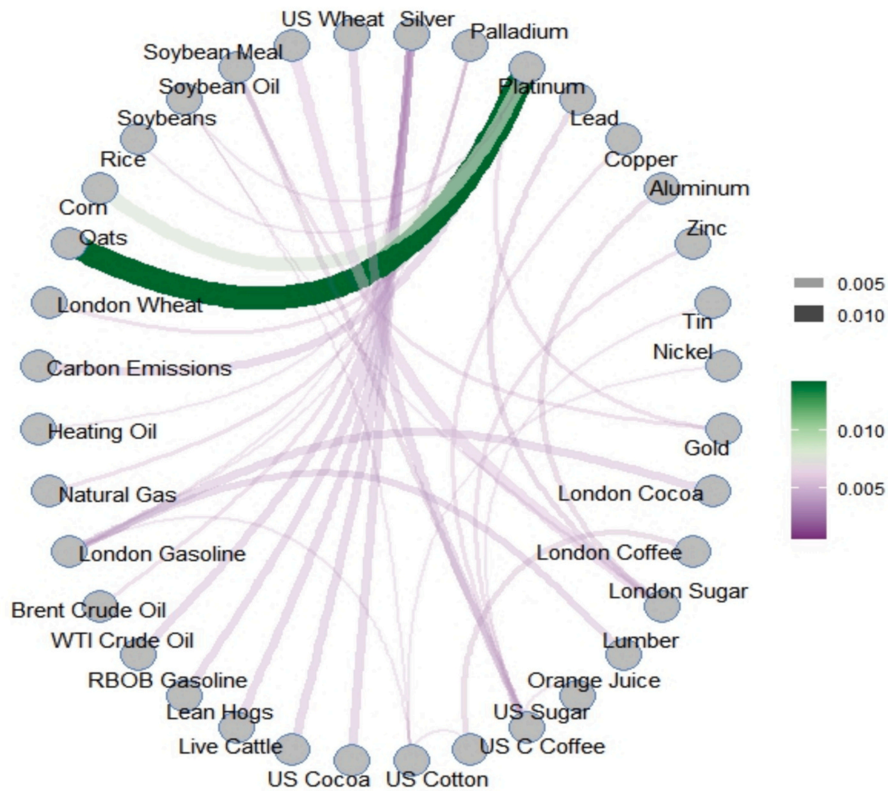

Fig. 2. Co-bubble network across the 36 commodity futures based on MST for the full sample period. Note: See notes to Fig. 1.

dependence. In the Grain and Cereal group, US Wheat and Rice have a relatively lower sensitivity to bubbles than other grain and cereal commodities, whereas Soybean Oil and Soybean have the strongest bubble sensitivity.

Interestingly, Fig. 4 illustrates the four co-bubble networks constructed based on MST. Compared to Fig. 2, it provides more granulated evidence of the bubble connectivity within each commodity group. The topology reveals the co-bubble dependency between the nodes within the commodity groups in the co-bubble networks. Lead, Platinum, Lean Hogs, London Cocoa, Natural Gas, and Rice are the core nodes in their respective groups and play an important role in the connectivity of the entire co-bubble network, whereas Lead and Platinum have the same importance, and similarly for Lean Hogs and London Cocoa. Fig. 4 shows the presence of relatively independent groups of nodes in the four co-bubble networks, such as Sliver, Lumber, and Oats. These results extend our understanding of the co-bubble dependency across commodity groups based on the co-bubble network.

3.5. Dynamics ranking of co-bubble influence: all 36 commodity futures

In this subsection, we examine the rolling-sample ranking of co-bubble centrality, which provides valuable insights into the time-evolution of co-bubble influence across the 36 commodity futures under study. Fig. 5 illustrates the co-bubble network centrality rankings using a colour gradient ranging from deep blue to deep red (the deep red (blue) reflects high (low) ranking based on co-jump network centrality), which reflects a specific ranking ranging from 1 to 36.

The ranking of co-bubble influence in the Energy group gradually changes over the full sample period. Before COVID-19 (2018 to 2019), the Energy group ranks highest in terms of co-bubble network centrality, especially Carbon Emissions and WTI Crude Oil. This suggests that Energy futures bubbles have a more significant impact on the bubbles of other commodity futures. Such a high susceptibility to bubbles in Energy futures prices can be attributed to several factors. Firstly, energy futures are significantly influenced by macroeconomic factors and notably news sentiment (Maslyuk-Escobedo et al. (2017), possibly resulting in price volatility with high correlation and contagion effects. Secondly, the energy futures constitute strategic commodities, such as crude oil, which often attracts large attention from traders and investors and thus an increased trading activity (Ji et al., 2020), further amplifying the impact of its price bubbles on the bubble formation of other commodity futures. Several related studies such as Bouri et al. (2021) and Shahzad et al. (2021) highlight the importance of energy as a strategic commodity group for the system of all commodity. Lastly, extreme events can burst price bubbles in the energy market, as reflected in their lower rankings during the Russo-Ukrainian conflict.

During the COVID-19 pandemic, the ranking of the Metals group in terms of co-bubble network centrality gradually improves, particularly for Gold and Palladium. Possible explanations include the following. Firstly, the metal futures market is perceived as a safe-haven asset during uncertain and volatile market environments, attracting increased attention and trading activity from investors and hedgers (Rehman et al., 2019). As a representative of this safe-haven asset, gold exhibits strong performance during market turmoil, thus the improvement in its ranking of co-bubble network centrality may reflect the increased investor demand for safe haven assets. Secondly, the COVID-19 pandemic has had widespread impacts on the global economy and financial markets, resulting in increased market uncertainty and volatility. In such circumstances, investors tend to pay closer attention to the metal market and incorporate it into their portfolios as a means of seeking refuge or generating better risk-adjusted returns (Dutta et al., 2020). This could contribute to the higher ranking of co-bubble network centrality in the metal futures market.

After the COVID-19 pandemic, the ranking of gold gradually decreases. Aluminium and copper demonstrate relatively higher rankings in terms of co-bubble network centrality, while carbon emissions maintains its dominant position in the centrality ranking during the Russo-Ukrainian conflict. Several factors can account for these

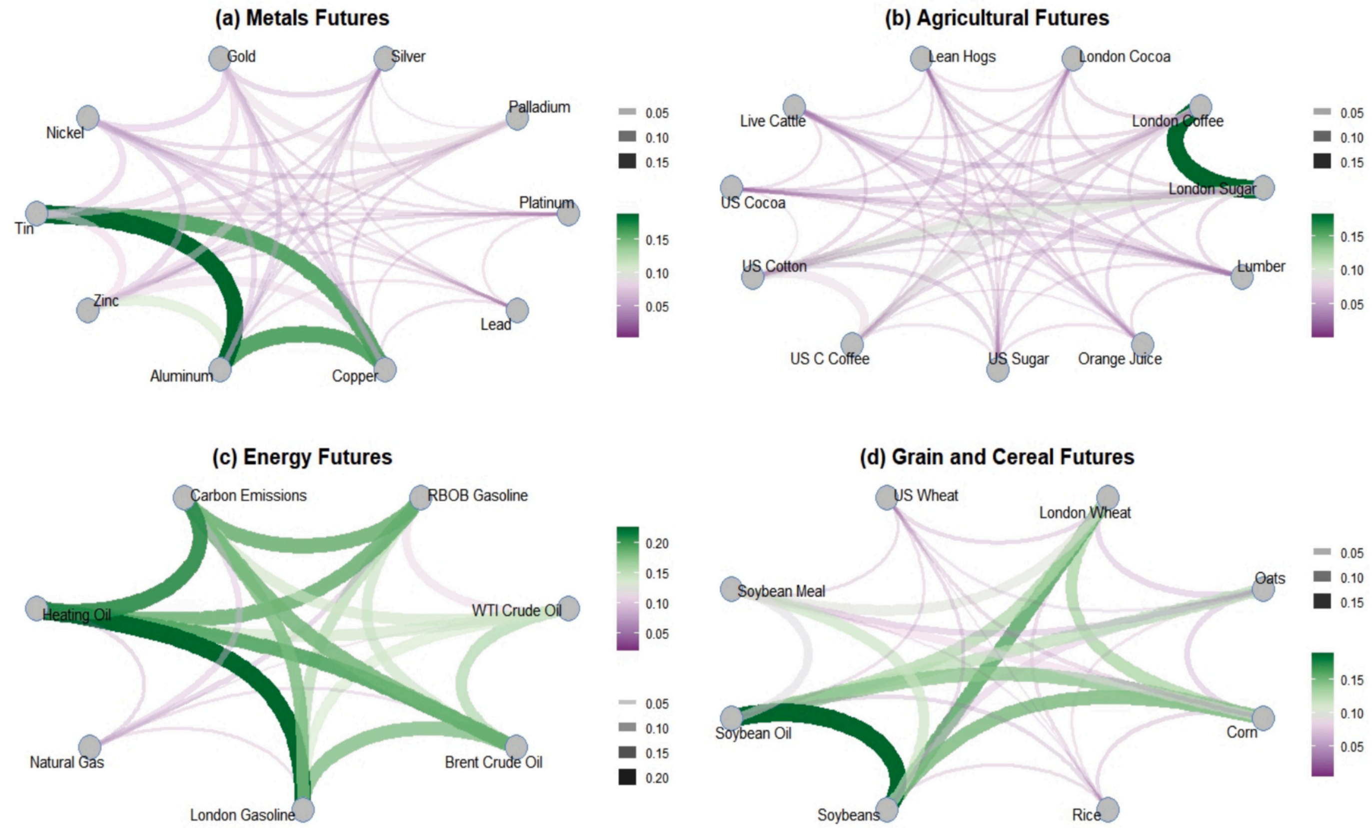

Fig. 3. Co-bubble networks in the commodity futures based on inter-industry groups for the full sample period. Notes: The edges connecting pairs of commodity futures represent the intensity of the co-bubble. In the colour bar (legend at bottom right), the colour distribution, ranging from deep green to deep purple, reflects the intensity of co-bubble relationships between pairs of commodity futures (deep green (deep purple) means higher (lower) intensity). In the colour bar (legend at top right), the colour distribution, ranging from light grey to black, reflects the width of the edge connecting the pairs of commodity futures, with grey (black) indicating a narrower (wider) width. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

observations. Firstly, with the easing of the pandemic and gradual global economic recovery, investor demand for safe-haven assets such as gold has diminished, possibly leading to a decline in the centrality ranking of gold. Secondly, aluminium and copper are widely used in industrial production, with their prices influenced by global economic conditions and industrial demand. Therefore, during periods of economic recovery, the demand for aluminium and copper tends to increase,[4] attracting more investor attention, which could result in higher centrality rankings in these industrial metals. Lastly, the Russo-Ukrainian conflict has triggered a spike in the levels of geopolitical risk and uncertainty in energy supply, leading to increased price volatility in the carbon emissions market (Wang et al., 2022). Notably, since the start of the war, many governments have reduced their reliance on natural gas and turned back to more polluting fossil fuels, leading to an increase in CO2 emissions, and ultimately to more demand on carbon emission allowances.[5] Furthermore, responsible investors incorporate carbon emissions into their portfolios as a means of seeking refuge or generating returns, elevating its centrality ranking in the co-bubble network.

3.6. Dynamics ranking of co-bubble influence: inter-commodity groups

We focus on the analysis of the dynamics ranking of co-bubble influence across all 36 commodity futures in Section 3.5. Here, we shift our focus to the dynamics ranking of co-bubble influence within each commodity group, which allows us to capture the dynamic co-bubble influence within the Energy, Metals, Agricultural, and Grain and Cereal commodity groups. Conducting an inter-group study has notable theoretical and practical advantages compared to a more comprehensive analysis. Firstly, by examining the co-bubble dependency and interaction effects among various commodity groups, we gain a better understanding of the dynamics associated with these interactions, enabling a more refined and accurate explanation of the impacts of price bubbles and co-bubbles across commodity groups across time. Secondly, leveraging the insights derived from the co-bubble impact between commodity groups, investors can develop improved trading strategies and optimize their portfolios for enhanced performance.

Fig. 6 presents the centrality rankings of co-bubble networks for the commodity groups, with a colour scheme ranging from deep blue to deep red (the deep red (blue) reflects high (low) ranking based co-jump network centrality). Interestingly, while gold does not consistently achieve an optimal centrality ranking in the comprehensive analysis of all commodity futures (see Fig. 5), it maintains a dominant position in the centrality ranking within the Metals group during the COVID-19 pandemic, possibly because of its safe-have property during crisis periods (Rehman et al., 2019; Dutta et al., 2020). This suggests that price bubbles in gold during crisis periods have a significant impact on other commodities in the Metals group, rendering them more vulnerable to the formation of gold bubbles. Similar patterns are observed in Tin and Soybean Oil during the Russo-Ukrainian conflict, where they also maintain prominent positions in the centrality ranking. In contrast, Natural Gas exhibits the lowest dynamic rankings within the Energy group. Several factors can explain this observation. Firstly, the natural

[4] Notably, copper is used in various industrial sectors.

[5] https://climate.ec.europa.eu/news-your-voice/news/eu-carbon-market-report-driving-emission-reductions-and-enabling-climate-and-energy-investment-2022-12-14_en

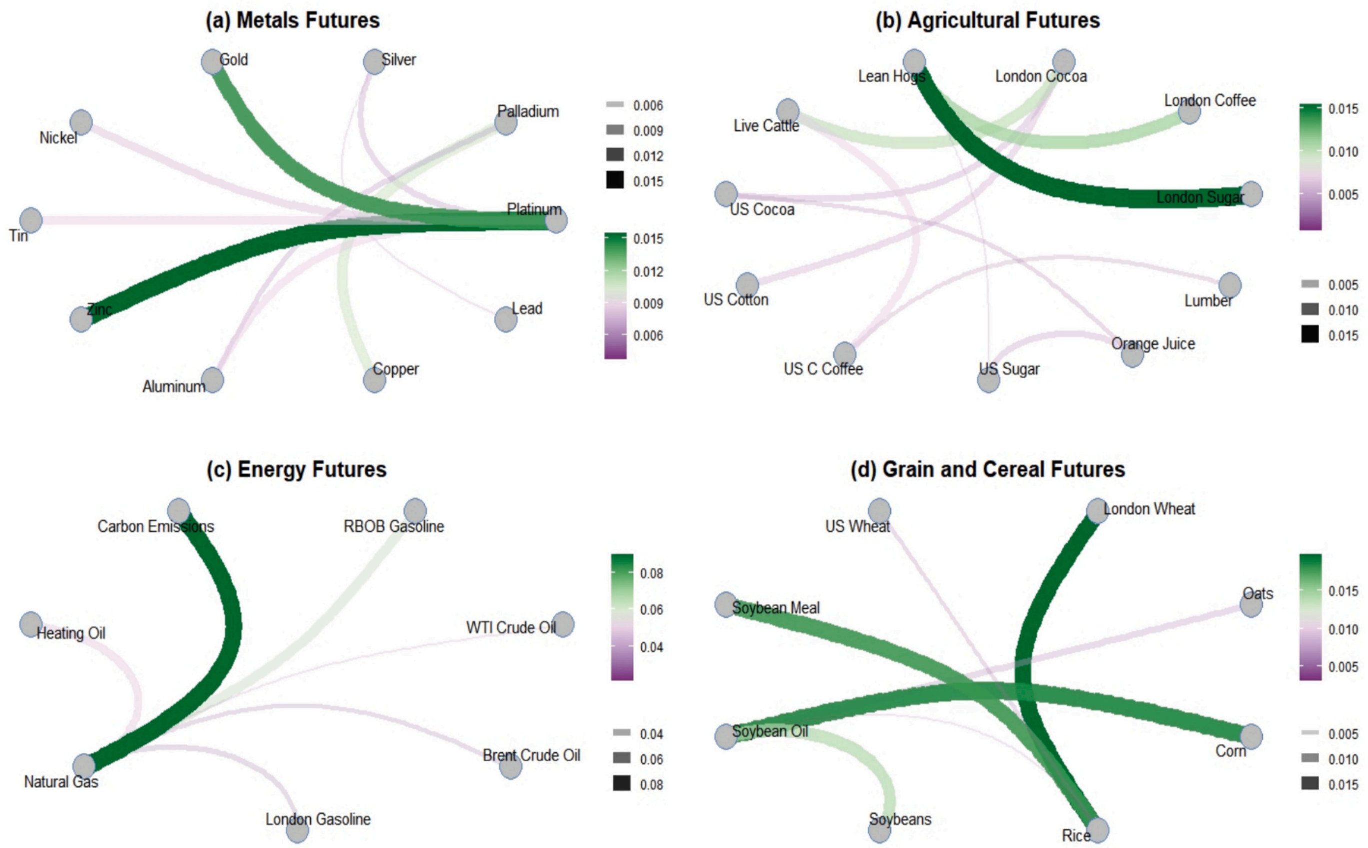

Fig. 4. Four co-bubble networks in the commodity futures based on MST for the full sample period. Note: See notes to Fig. 3.

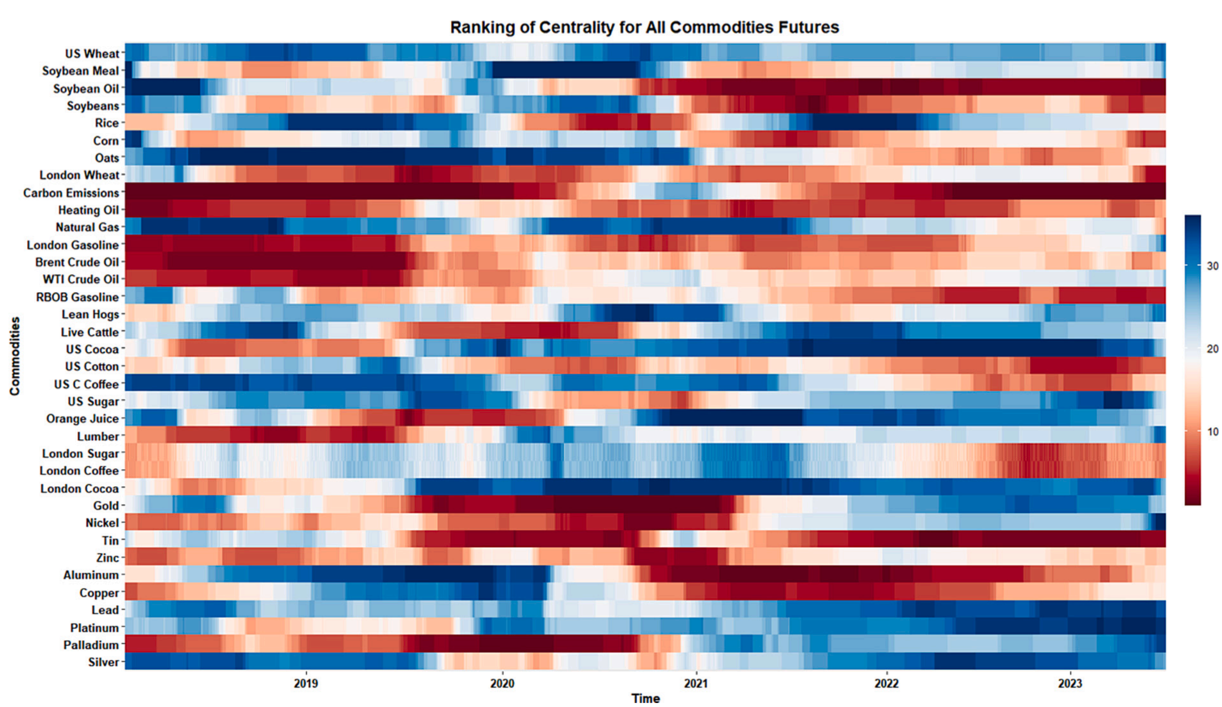

Fig. 5. Dynamic ranking of co-bubble influence of the 36 commodity futures. Notes: In the colour bar (legend on the right), each colour stands for a ranking of co-bubble influence, varying from 1 to 36. The deep red (blue) reflects high (low) ranking based co-jump network centrality. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

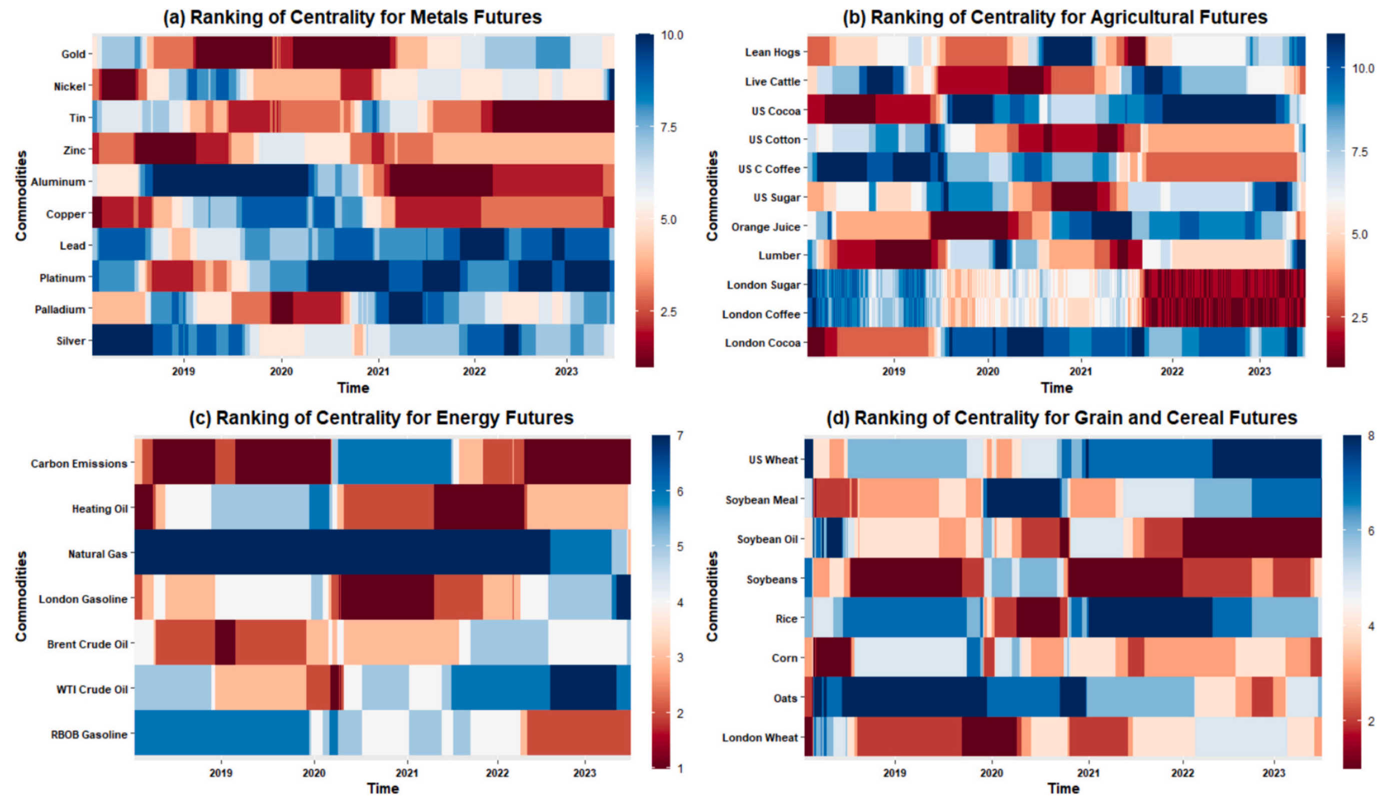

Fig. 6. Dynamic ranking of co-bubble influence in the commodity futures for the four industry groups. Note: In the colour bar (legend), each colour stands for a ranking of co-bubble influence.

gas market operates differently compared to other energy commodities, given it is often regionally fragmented due to transportation constraints and infrastructure limitations. This fragmentation can limit the spillover effects of price bubbles from natural gas to other commodities within the Energy group. Secondly, natural gas exhibits strong seasonality in demand, which can dampen the formation of price bubbles and reduce the overall impact on other commodities in the group. Finally, natural gas has a diverse supply base, which can mitigate the potential for price bubbles and limit their contagion effects on other energy commodities. However, during the COVID-19 pandemic, London Gasoline emerges as the dominant commodity in the Energy group, while Carbon Emissions continue to maintain their dominant position during the Russo-Ukrainian conflict. Furthermore, it is worth noting that there are strong co-bubble interaction effects between London Sugar and London Coffee throughout the sample period, indicating a significant relationship between the price bubbles of these two agricultural commodities.

4. Application to portfolio allocation

Investors and portfolio managers are concerned with the construction of optimal investment portfolios that can maximize their returns while restraining the level of risk. To showcase the economic value of the co-bubble network in commodity futures markets, we devise several daily rebalanced mean-variance strategies tailored to the specific requirements of various investors. In our portfolio analysis, we distinguish between two investment scenarios: one for investors solely focused on a specific commodity group and another one for investors interested in the overall commodity markets. This differentiation allows us to address the various needs and objectives of these two clusters of investors. By leveraging the dynamic ranking of co-bubble influence for each trading day throughout the study period, we aim to demonstrate how these strategies can effectively guide investors constructing portfolios that align with their investment goals and preferences.

4.1. Mean-variance portfolio construction

To validate the effectiveness of portfolio strategies, it is imperative to select a set of classical investment portfolio strategies as benchmarks to compare with the proposed strategy. Conventional portfolio models typically construct optimization objectives for portfolio construction based on asset returns and volatilities. Subsequently, historical return sequences are employed to estimate the parameters of the investment portfolio. Consequently, we can formulate mean-variance portfolio (MV) weights w_t, on day t, by solving the following constrained optimization equation:

$$\begin{aligned} &\min_{w_t} w_t^T \widehat{\Sigma}_{t-1} w_t \\ &\textit{subject to } w_t^T \overrightarrow{1} = 1 \\ &\|\mathrm{w_t}\|_1 \leq l, \end{aligned} \tag{15}$$

where $\widehat{\Sigma}_{t-1}$ is the covariance matrix, and $l \geq 0$ is the level of leverage.

The daily mean-variance portfolio return can be calculated as:

$$r_t^{MV} = w_t^T r_t \tag{16}$$

where $\boldsymbol{r_t}$ denotes a vector of commodity futures logarithmic returns.

The Sharpe ratio is a commonly used metric for evaluating portfolio performance, providing a measure of the trade-off between portfolio returns and volatility. A value higher than zero indicates that the portfolio's returns exceed the risk-free rate, while a negative value suggests that the portfolio's returns fall below the risk-free rate. Furthermore, a higher positive value of the Sharpe ratio signifies a higher excess return for the portfolio for the amount of risk taken. Therefore, we can employ out-of-sample Sharpe ratios as benchmarks to compare various portfolio strategies. The annualized Sharpe ratio (Sharpe, 1998) is calculated as:

$$sr^{MV} = \frac{\overline{r^{MV}} \times 300}{\sigma^{MV}} \tag{17}$$

where $\overline{r^{MV}}$ is the average daily return of the mean-variance portfolio, and σ^{MV} represents the annualized volatility of the mean-variance portfolio.

4.2. Analysis of portfolio performance

In this empirical portfolio analysis, we consider various mean-variance portfolio strategies and leverage constraints for optimization purposes. The portfolio strategies are based on dynamic ranking of the co-bubble influence across 36 commodity future markets. It is important to note that all our analyses are conducted out-of-sample (period spanning September 19, 2017 to July 1, 2023), as we calculate the mean-variance portfolio weights traded on day $t + 1$ using covariance estimates from day t, employing a rolling window of size 300 days.

Table 3 presents the annualized Sharpe ratios for various mean-variance portfolios, allowing for comparison after adjusting for portfolio returns. Notably, the co-bubble dependency-driven portfolio outperforms the baseline mean-variance portfolio. Analysing the ranking of co-bubble influence across the 36 commodity future markets, portfolios constructed using the top 10 commodity futures, based on centrality, exhibit the highest Sharpe ratios compared to other mean-variance portfolios. However, when the number of commodities is decreased to 5, there is a deterioration in the Sharpe ratio,[6] whereas portfolios built on the top 15 and top 20 commodity futures exhibit higher Sharpe ratios compared to the top 5 portfolio, but lower Sharpe ratios compared to the top 10 commodity futures.

As leverage constraints are relaxed, all mean-variance portfolios present an increasing annualized Sharpe ratio. We offer explanations for these findings. The selection of the top 10, 15, and 20 commodity futures, which hold significant influence on the price bubble behaviour of other commodity futures while being less susceptible to such behaviour themselves, implies their relative bubble independence from other commodity futures prices. Consequently, when combined into a portfolio, these commodity futures achieve improved risk diversification, resulting in a reduction in portfolio volatility. Conversely, commodity futures with lower centrality are more affected by other commodity futures, which make them less useful for diversification purposes. It is noteworthy that when utilizing the top 5 commodity futures, the co-bubble dependency among commodity futures may be underestimated, leading to increased portfolio volatility, resulting in a lower Sharpe ratio.

Table 3
Annualized Sharpe ratio of mean-variance portfolios.

Leverage	Baseline-MV	Top5 Rank-MV	Top10 Rank-MV	Top15 Rank-MV	Top20 Rank-MV
1	0.9223	0.5170	**1.1235**	1.0272	0.9708
2	1.1182	0.6357	**1.2691**	1.1853	1.1433
3	1.1835	0.6753	**1.3177**	1.2380	1.2007
4	1.2162	0.6951	**1.3419**	1.2643	1.2295
5	1.2358	0.7069	**1.3565**	1.2802	1.2467
6	1.2488	0.7148	**1.3662**	1.2907	1.2582
7	1.2582	0.7205	**1.3731**	1.2982	1.2664
8	1.2652	0.7247	**1.3783**	1.3039	1.2726
9	1.2706	0.7280	**1.3824**	1.3083	1.2774
10	1.2750	0.7306	**1.3856**	1.3118	1.2812

Notes: This table displays the annualized Sharpe ratios of mean-variance portfolios under various constrained leverages. The Baseline-MV strategy represents a portfolio construction approach that considers all commodity futures to construct optimal investment portfolios. The Top5 Rank-MV strategy, on the other hand, focuses on the top 5 commodity futures based on the ranking of co-bubble influence. Similarly, the Top10 Rank-MV, Top15 Rank-MV, and Top20 Rank-MV strategies consider the top 10, top 15, and top 20 commodity futures, respectively, in terms of their ranking of co-bubble influence.

Fig. 7 illustrates the trajectory of portfolio gains by plotting the cumulative returns of multiple mean-variance portfolio strategies over time. The portfolios constructed using the co-bubble network model outperform the baseline strategy by achieving higher profits, while also experiencing lower volatility and shorter downside periods. Among these investment portfolios, the Top10 Rank-MV strategy, which considers the top 10 commodity futures based on the ranking of co-bubble influence, exhibits more substantial profit differentials. Importantly, following periods of crisis such as the COVID-19 pandemic or the Russo-Ukrainian conflict, the Top10 Rank-MV portfolio rapidly expands its profit differential. However, as the portfolio includes a higher number of commodity futures exceeding 10, such as 15 and 20, the profit differential gradually diminishes and may even fall below the benchmark strategy during some periods, such as the COVID-19 pandemic. We provide explanations for these findings. Firstly, the selection of the top 10 commodity futures based on their co-bubble influence implies their relative resilience to the price bubble behaviour of other commodity futures. This characteristic contributes to higher profits for the Top10 Rank-MV portfolio compared to other portfolios, particularly in the aftermath of crisis periods. Secondly, after crisis periods, certain commodities may experience significant price bubbles due to supply disruptions, changes in demand, or geopolitical factors. The Top5 Rank-MV portfolio effectively captures these return bubbles, leading to increased profitability. Finally, when the number of commodity futures in the portfolios is reduced, as in the case of the Top5 Rank-MV portfolio, the commodity portfolios may exhibit higher correlation and a higher tendency to move together, resulting in heightened volatility and thus lead to a decrease in the Sharpe ratio.

5. Conclusion

In this study, we first examine the presence of bubbles in 36 commodity futures price series via the BSADF-based procedure, which allows us to identify the start and end of each commodity futures price bubble period and then propose and construct co-bubble networks to model the interdependency structure among commodity futures markets. Our approach sheds light on the co-bubbling transmission across 36 commodities and various inter-commodity groups. Interestingly, the proposed co-bubble network offers a fresh perspective on the interrelationships and dependencies among commodity futures price bubbles. Importantly, we analyse the portfolio implications of these bubble dynamics, providing valuable insights into the performance of portfolios constructed based on the bubbles of commodity futures.

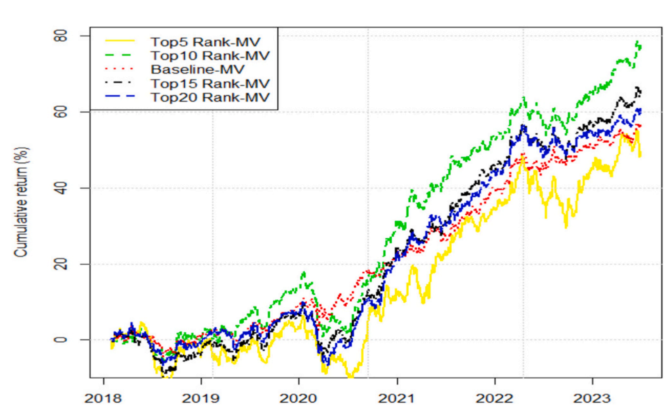

Fig. 7. Cumulative returns of the various mean-variance portfolio strategies.

[6] Portfolios utilizing the top 5 commodity futures demonstrate lower Sharpe ratios in comparison to the baseline mean-variance portfolio.

Our findings reveal that commodity futures within the Energy group exhibit a higher likelihood of experiencing co-bubbles. However, the analysis of co-bubble networks, based on inter-commodity groups, shows that co-bubble heterogeneity exists among pairs of commodity futures prices within these groups. The influence of co-bubble centrality is time-varying, indicating its sensitivity to crisis periods. Specifically, we reveal the dynamics of co-bubble influence rankings during the COVID-19 pandemic and the Russo-Ukrainian war. During the pandemic, a gradual improvement emerges in the ranking of the Metals group in terms of co-bubble network centrality, particularly for Gold and Palladium. During the Russo-Ukrainian war, Carbon Emissions maintain its dominant position in the centrality ranking.

The above findings are important for policymakers who are particularly concerned with price and market stability. In this regard, the transmission of co-bubbles in the commodity markets might undermine food price stability and worry importers of food commodity staples, which require a particular attention from both commodity importers and exporters. Furthermore, the detailed evidence of co-bubble transmission in the commodity markets should be very informative for policymakers given that a contractionary monetary policy can effectively ease asset price bubbles.[7]

For investors and portfolio managers, the presence of co-bubbles in various commodity groups suggests that the co-bubble behaviour of commodity prices differs across these groups. This highlights the need to account for the unique characteristics of each commodity group when modelling the dynamic bubble behaviour of commodity futures. To further assess the practical implications of our co-bubbles network analysis, we evaluate the portfolio implications of the significant changes in co-bubble dependencies, which should concern investors and portfolio managers. The mean-variance portfolio strategies based on the dynamic ranking of co-bubble influence yield a higher Sharpe ratio compared to the baseline portfolio strategy. They underscore the importance of considering not only the presence of co-bubbles but also the temporal co-bubble behaviour of commodity futures prices and the impact of specific commodities on others.

Future research could expand our current findings in several ways. One possible research avenue concerns the co-bubble impact between stocks and commodity futures within a network model, along with its implications for portfolio management. This would provide a more comprehensive understanding of the interdependency and dynamic bubble behaviour between these two asset classes under various crisis periods. Another possible research avenue involves how to build trading strategies upon the highly interdependent co-bubble commodity pairs in order to obtain abnormal returns under a weak-form of market efficiency.

CRediT authorship contribution statement

Yan Chen: Conceptualization, Data curation, Formal analysis. **Elie Bouri:** Investigation, Writing – original draft, Writing – review & editing. **Lei Zhang:** Conceptualization, Data curation, Methodology, Supervision, Writing – original draft, Writing – review & editing.

Acknowledgments

This study is partially supported by the MOE Layout Foundation of Humanities and Social Sciences (Grant No. 22YJA790007), and the Fundamental Research Funds for the Central Universities.

Appendix A. Appendix

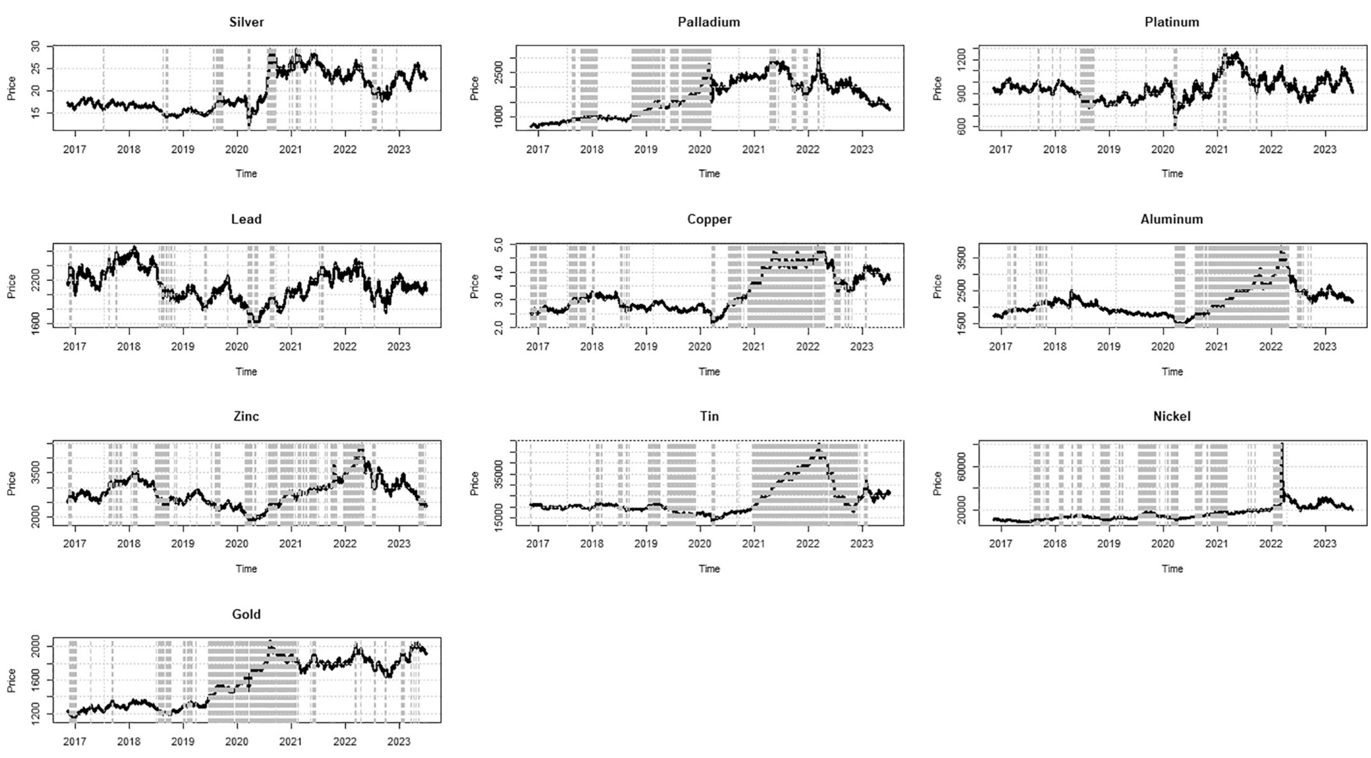

Fig. A1. GSADF: Explosive behaviour periods in the Metals group.

[7] As indicated by Chan et al. (2024) for green and brown sectors.

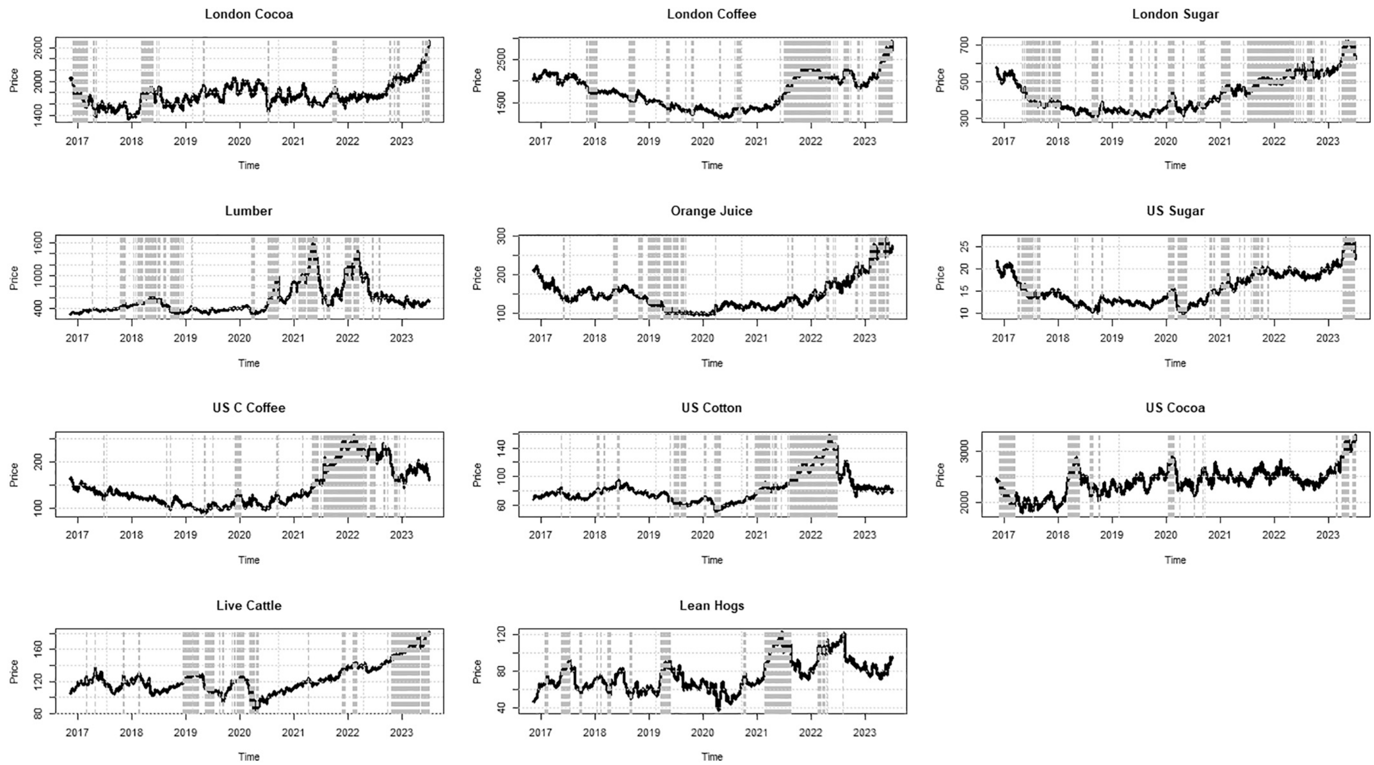

Fig. A2. GSADF: Explosive behaviour periods in the Agricultural group.

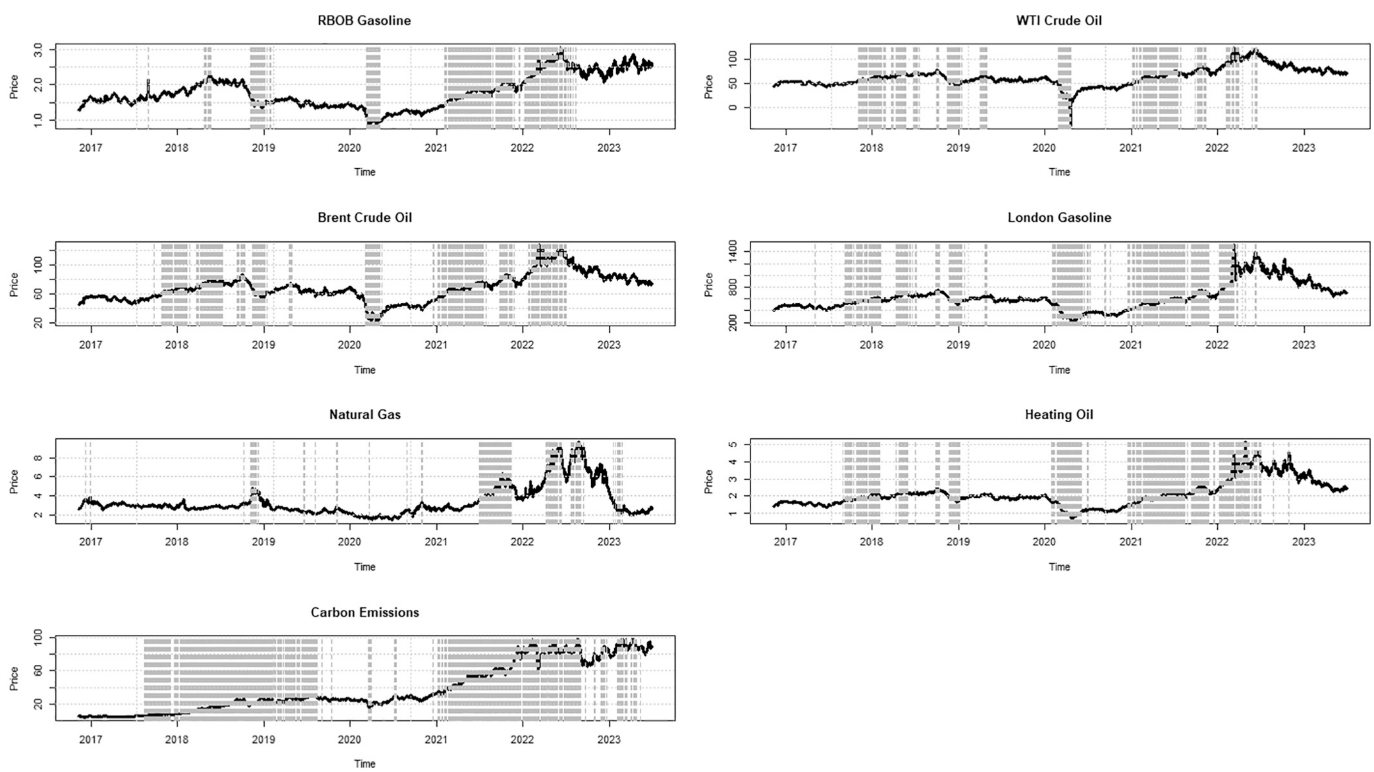

Fig. A3. GSADF: Explosive behaviour periods in the Energy group.

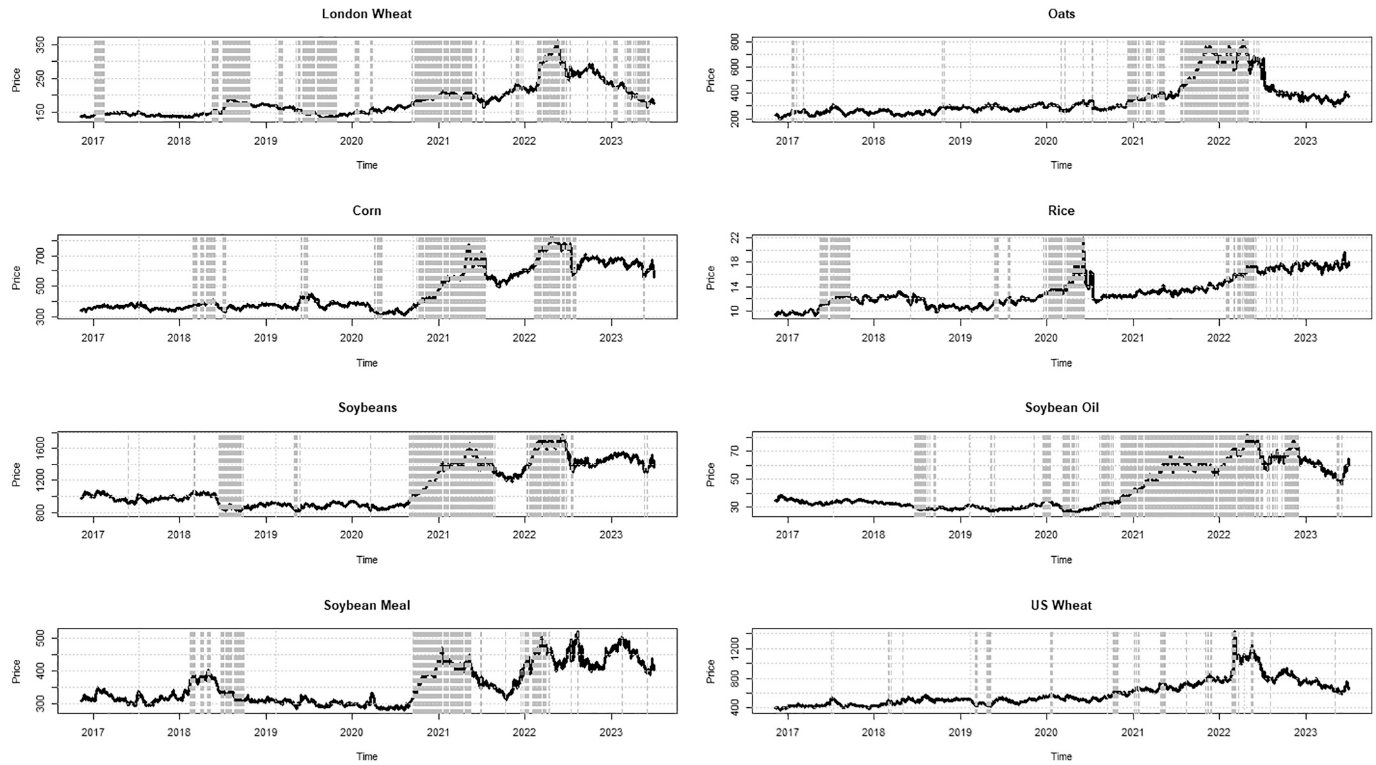

Fig. A4. GSADF: Explosive behaviour periods in the Grain and Cereal group.

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eneco.2025.108839.

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